Distributional Curvature and Heat-Index Budgets

A Scoped Ram Analysis of Gigli's Alexandrov Characterization

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Abstract

The assigned manuscript asserts an equivalence between an Alexandrov lower curvature bound and Gigli's global distributional sectional inequality on a noncollapsed finite-dimensional RCD space. Its heat-index proof combines an exact inverse-viscosity cancellation with a positive, potentially singular Bochner measure. We retain that structure and give an explicit finite-viscosity acceleration budget, including the nonnegative completed-square and osmotic-Hessian defects. A physical coefficient-tail modulus replaces an unspecified small error; additional L^p control of the fixed trial derivatives gives conditional rates, whereas L^2 alone does not. Weak Hessian measurements and manufactured Gaussian identities illustrate the scope of finite checks. The global transport, frame, companion Hessian-comparison, and Alexandrov-globalization arguments remain upstream inputs, not consequences of these local estimates. Contribution assessment: C2/5, the contribution is a modest quantitative account of the source's acceleration cancellation and its necessary coefficient and singular-measure controls, not a new curvature characterization.

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1 Model, claim, and source status

Write $\mathfrak{m} = \mathcal{H}^n$, $n \geq 2$, and fix $\kappa \in \mathbb{R}$. The source [1] works on a complete separable metric space with full-support reference measure and the *unreduced* $\mathrm{RCD}((n-1)\kappa, n)$ condition. Its claimed equivalence is with $\mathrm{CBB}(\kappa)$, under the global distributional inequality

$$\mathcal{R}(X,Y,Y,X)(f) \geq \kappa \int f(|X|^2|Y|^2 - \langle X,Y\rangle^2)\, d\mathfrak{m}. \tag{1}$$

Here X, Y range over the source's global test vector fields and f over nonnegative global test functions. This is not a statement merely on the regular set, nor a finite list of tensor entries.

The Laplacian is the nonpositive L^2 generator: $P_s = e^{s\Delta}$. The test algebra consists of bounded globally Lipschitz representatives in $D(\Delta)$ with $\Delta f \in W^{1,2}$; the smaller Test^∞ class additionally has bounded Laplacian. These are Sobolev test classes, not an assertion of classical smoothness. A test vector field is a finite sum $a\nabla b$ with a, b in the test algebra. It has L^2 covariant derivative and divergence and is bounded. The source extends tests by compact localization and suitable strong graph-norm approximation; on an infinite-measure space a coefficient identically one need not be a global test function.

The convention is $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$. In particular its distributional realization is

$$\begin{aligned}\mathcal{R}(X,Y,Z,W)(f) = &-\int \langle \nabla_X(fW), \nabla_Y Z\rangle\, d\mathfrak{m} - \int f\langle \nabla_Y Z, W\rangle \operatorname{div} X\, d\mathfrak{m}\\ &+\int \langle \nabla_Y(fW), \nabla_X Z\rangle\, d\mathfrak{m} + \int f\langle \nabla_X Z, W\rangle \operatorname{div} Y\, d\mathfrak{m}\\ &-\int f\langle \nabla_{[X,Y]}Z, W\rangle\, d\mathfrak{m}.\end{aligned} \tag{2}$$

All contractions below use this physical tangent-module metric and the actual measure, not a Euclidean coefficient norm substituted for it.

Public status. Gigli's 2019 paper introduces this distributional tensor and explicitly poses the Alexandrov characterization as a conjecture [2]. Brena–Gigli describe measure-valued tensor representations [3]; that abstract is not a replacement for the representation theorems used here. The 2026 published theorem of Erös–Kunzinger–Ohanyan–Vardabasso gives local Alexandrov comparison from distributional lower sectional bounds for C^1 Riemannian metrics on a smooth manifold [4]. Its domain is different from the general RCD assertion under study. The external refresh inspected the first two abstracts/version records and the latter article's main statement and metadata, not all external proofs.

The assigned September 24 manuscript is analyzed as the pinned repository text. Equality with the captured October 8 tree is byte provenance, not a proof or a claim about subsequent live changes. Reading its complete scientific text does not independently establish every imported theorem.

2 Architecture and unreplaced dependencies

The proof has two directions and several limits that cannot be reordered by a finite weak measurement.

Alexandrov to distributional curvature. The source's forward module starts from finitely supported sign-symmetric laws of compact test gradients. Regular Lagrangian flows have velocities $hA + h^2 r S_A$, with S_A fixed while $h \downarrow 0$. Endpoint densities have a first weak expansion; their pairwise optimal interpolations have a common compact support, uniform density bounds, and density-weighted strong convergence of rescaled velocities. A constant-curvature cosine expansion supplies the fourth-order geometric term. Optimal-transport duality and a completed square identify the curvature coefficient. Endpoint corrections then approach $-\nabla_A A$ in L^2, after the h limit.

Finite gradient choices, weighted partitions, and sign averaging recover the sectional test. The subsequent passage to variable coefficients uses positive distributions and grouped polynomial-coefficient signed measures. It does not evaluate arbitrary tangent fields on a singular support. Compact approximations and a final exhaustion recover global tests. Uniform bounds for derivatives of all partition functions are not claimed: their relevant appearances cancel before the limiting step.

Distributional curvature to a weak Hessian bound. The heat-index module constructs a positive heat factorization, cancels all dangerous $1/c$ terms, and treats the remaining positive Bochner measure using quasi-continuity and a one-sided Laplacian bound. The zero-viscosity passage uses action equality, not merely weak velocity convergence. This produces a common terminal path prescription, then allows removal of the auxiliary positive terminal density.

The frames module separately supplies all-time frames outside a capacity-zero set and a rank-defect set of zero $\mathcal{H}^{n-1}$ measure. Almost-everywhere regularity alone would not exclude exceptional times on a path. If P is the Gram matrix of a finite frame along a path and C its connection matrix, the identities $P' = C^T P + PC$ and $a' + Ca = 0$ give parallel transport as an isometry for $a^T P a$. Inverse Gram bounds are path dependent. Countable trial families are dense pathwise in the appropriate $W^{1,2}$ space; this is stronger and more specific than a finite matrix rank check.

The final comparison is a genuine external interface. For $q = d(p,\cdot)^2/2$ the source obtains the model radial Hessian bound with nonpositive singular part. A bounded modification $F = \chi(q)$, constant outside a ball, has

$$H_F(Z,Z)(h) = H_q(Z,Z)(h\chi') + \int h\chi''\langle \nabla q, Z\rangle^2 \, d\mathfrak{m}. \tag{3}$$

The companion weak-Hessian comparison theorem converts tests into an inequality along *every* minimizing geodesic, including on possibly collapsed RCD spaces. On the inner ball this gives $y'' + \kappa\ell^2 y \le \ell^2$ for the modified distance along a segment of length ℓ. A Dirichlet comparison, local triangle gluing, and complete-space globalization finish the claimed route. Our report does not reprove that companion theorem, all-time frame existence, or globalization. In particular a rate for one heat-index error is not a rate for Alexandrov comparison.

3 The physical heat-index identity

Fix $0 < \delta < 1$, viscosity $c > 0$, and the source's positive terminal probability density f. Before normalization it has the form $f_0 + \lambda e^{-D_0 d(o,x)}$, with compactly supported Lipschitz $f_0 \ge 0$, $\lambda > 0$, and D_0 sufficiently large. Its logarithm is Lipschitz for each fixed λ; constants need not be uniform as $\lambda \downarrow 0$. The initial datum u is bounded and globally Lipschitz and constant outside a compact set. Put

$$\begin{aligned} &H_t = P_{ct}(e^{-u/(2c)}), && J_t = P_{c(1-t)}(f/H_1), && \rho_t = H_t J_t, \\ &\phi_t = -2c\log H_t, && \psi_t = 2c\log J_t, && b_t = (\psi_t - \phi_t)/2, \quad \theta_t = (\psi_t + \phi_t)/2. \end{aligned} \tag{4}$$

Let $v = \nabla\theta$, $w = \nabla b$, and $q = \Delta b$. Then

$$\partial_t \rho = -\operatorname{div}(\rho v), \qquad \nabla\rho = \rho w/c, \qquad \partial_t\theta = -\tfrac{1}{2}(|v|^2 + |w|^2) - c\Delta b, \qquad \rho_1 = f. \tag{5}$$

Trials are $Y_t = \nabla g_t = \sum_j a_j(t)\nabla g_j$, with a fixed finite collection of compact test functions and smooth time coefficients, $Y_\delta = 0$, $Y_1 = Z$. Choose a compact region E containing their supports. All time-space norms in this report are on $\Omega = (\delta, 1) \times E$, unless specified otherwise.

For a locally $W^{1,2}$ scalar F, its weak Hessian test is

$$H_F(Y,Y)(\xi) = -\int \operatorname{div}(\xi Y)\langle \nabla F, Y\rangle \, d\mathfrak{m} - \int \xi\langle \nabla F, \nabla_Y Y\rangle \, d\mathfrak{m}. \tag{6}$$

This is meaningful without an L^2 Hessian of F. Set $\mathcal{D}_c Y = \partial_t Y + \nabla_v Y$. The source proves, in the time-distribution sense and then integrated,

$$\begin{aligned} \frac{d}{dt}\int \rho H_\theta(Y,Y)\, d\mathfrak{m} = & \int \rho\big(2H_\theta(\mathcal{D}_c Y, Y) - |H_\theta Y|^2\big)\, d\mathfrak{m} \\ & - \mathcal{R}(Y, v, v, Y)(\rho) - \mathcal{A}_c(Y), \\ \mathcal{A}_c(Y) = & H_{|w|^2/2+cq}(Y,Y)(\rho). \end{aligned} \tag{7}$$

The notation for curvature with weight ρ means the source's compactly localized admissible extension. No global L^2 membership of a constant heat datum is inferred.

3.1 The exact residual and its singular term

Write $d_Y = \operatorname{div} Y$ and $A_Y = \nabla_Y Y$. After the inverse-c cancellation, the source's identity is

$$\mathcal{A}_c(Y) = \int \rho |H_b Y|^2 \, d\mathfrak{m} + \mathcal{R}(Y, w, w, Y)(\rho) + I_c(Y). \tag{8}$$

The exact residual is

$$\begin{aligned} I_c(Y) = & -2\int \rho H_b(\nabla_w Y, Y)\, d\mathfrak{m} \\ & + 2\int \rho q(\langle w, A_Y\rangle + (Yb)d_Y)\, d\mathfrak{m} \\ & + \frac{1}{c}\int \rho\big[-2(Yb)\langle w, \nabla_w Y\rangle + |w|^2(Yb)d_Y + |w|^2\langle w, A_Y\rangle\big]\, d\mathfrak{m} \\ & + \int \rho\, cq\, dM_t + c\int \rho q\, \ell_t \, d\mathfrak{m}. \end{aligned} \tag{9}$$

Here $K = (n-1)\kappa$ and

$$M_t = d_Y^2\mathfrak{m} + \Gamma_2(g_t) - K|Y|^2\mathfrak{m} \geq 0, \qquad \ell_t = 2\langle Y, \nabla\Delta g_t\rangle + K|Y|^2. \tag{10}$$

The source only needs $d_Y \in L^2$, not L^4. The Γ_2 term may have a singular part; replacing M_t by its volume density would change the identity.

For the fixed finite catalogue the measures have a finite common dominating measure, ignoring capacity-zero sets. The scalar factors in the last line of (9) have quasi-continuous representatives. The source obtains a one-sided estimate

$$cq_t \geq -e_c(t) \quad \text{quasi-everywhere}, \qquad \int_\delta^1 e_c(t)\, dt = O_\delta(c|\log c| + c). \tag{11}$$

It first proves identities on closed interior time intervals and then passes to $T \uparrow 1$ at fixed c. This is essential: near the terminal time, pointwise derivative bounds blow up although the displayed error is integrable. Volume L^2 estimates cannot replace (11).

4 An explicit acceleration ledger

The next result makes the source's small error quantitative for one fixed catalogue and trial. Its hypotheses include the exact identities above and their terminal traces; it does not establish those identities by finite data. Let $d\nu_c = \rho\, dt\, d\mathfrak{m}$ on Ω, and suppose certified bounds are available for

$$\rho \le D, \quad |w| \le L, \quad |Y| \le B, \quad V = \nu_c(\Omega), \quad M_* = \sup_t M_t(E) < \infty, \tag{12}$$

$$H = \|H_b\|_{L^2(\nu_c)}, \quad Q = \|q\|_{L^2(\nu_c)}, \quad W = \left(\int_\Omega |w|^4/c^2\, d\nu_c\right)^{1/2}, \quad E_\ell = \|\ell\|_{L^2(\nu_c)}.$$

The density bound is used also for its quasi-continuous representative in the M_t integral. This follows in the source from its regularized density; it is not a license to choose an arbitrary version on a volume-null set. Since ρ_t has total mass one, $V \le 1 - \delta$. Put

$$T = |\nabla Y| + |d_Y|, \qquad J_c(T) = \int_\Omega |w|^2 T^2\, d\nu_c, \qquad J_c(1) = \int_\Omega |w|^2\, d\nu_c, \qquad e = \int_\delta^1 e_c(t)\, dt. \tag{13}$$

T is a fixed physical $L^2(dt\, d\mathfrak{m})$ function for the trial.

Theorem 4.1 (Finite-viscosity budget)**.** *Under* (1) *and the preceding fixed-trial hypotheses,*

$$\int_\delta^1 \mathcal{A}_c(Y)\, dt \ge \int_\Omega |H_b Y|^2\, d\nu_c - \mathcal{E}_c, \tag{14}$$

where $\kappa_- = \max(0, -\kappa)$ *and*

$$\mathcal{E}_c = B(2H + 2Q + 3W)\sqrt{J_c(T)} + \kappa_- B^2 J_c(1) + DM_* e + cQE_\ell. \tag{15}$$

Consequently, with terminal weak pairing as in (6)*,*

$$\begin{aligned} H_{\phi_1 + c\log f}(Z, Z)(f) + \int_\Omega |\mathcal{D}_c Y - H_\theta Y|^2\, d\nu_c + \int_\Omega |H_b Y|^2\, d\nu_c \\ \le \int_\Omega \left(|\mathcal{D}_c Y|^2 - \kappa |Y \wedge v|^2\right) d\nu_c + \mathcal{E}_c. \end{aligned} \tag{16}$$

All constants are attached to the actual heat factorization and fixed trial, not merely a finite compression.

Proof. The bounds $|\nabla_w Y| \le |w||\nabla Y|$, $|A_Y| \le B|\nabla Y|$, and $|Yb| \le B|w|$ imply that the first line of (9), integrated in time, has absolute value at most $2BH\sqrt{J_c(T)}$. The second line costs at most $2BQ\sqrt{J_c(T)}$. In the third line the absolute integrand is bounded by $3B\rho|w|^3 T/c$; Cauchy–Schwarz gives $3BW\sqrt{J_c(T)}$. The curvature term in (8) is bounded below by $-\kappa_- B^2 J_c(1)$.

Positivity of M_t, (11), and the representative-level density bound give $\int_\delta^1 \int \rho c q\, dM_t\, dt \ge -DM_* e$. The last volume integral costs at most cQE_ℓ. This proves (14) without discarding the positive osmotic-Hessian term. Finally

$$2H_\theta(\mathcal{D}_c Y, Y) - |H_\theta Y|^2 = |\mathcal{D}_c Y|^2 - |\mathcal{D}_c Y - H_\theta Y|^2.$$

Integrate (7), use $Y_\delta = 0$ and $\theta_1 = \phi_1 + c\log f$, and apply the sectional bound to its curvature term. First work up to $T_0 < 1$, then pass to $T_0 \uparrow 1$ using the assumed source endpoint and integrability estimates. The nonnegative defects can be retained by monotone convergence of their integrals. □

Remark 4.2. The source already contains the exact cancellation, positive-measure split, and completed-square identity. The increment in Theorem 4.1 is the explicit joint error ledger and its coefficient-tail consequences below. We do not assert strong convergence of H_θ, or pass either retained square through $c \downarrow 0$ without new compactness information.

5 Coefficient tails and conditional rates

The source obtains convergence with fixed L^2 trial derivatives. The following formulation says precisely which discarded part must be bounded if a finite coefficient calculation is used.

Proposition 5.1 (Tail modulus)**.** *Under* (12)*, for every cutoff* $a > 0$*,*

$$J_c(1) \le cW\sqrt{V}, \tag{17}$$
$$J_c(T) \le a^2 cW\sqrt{V} + DL^2 \int_{\{T>a\}} T^2 \, dt \, d\mathfrak{m}.$$

The infimum of the right side over $a > 0$ *is a valid bound. If* D, L, W, V *stay bounded and* T *is fixed in physical* L^2*, this infimum tends to zero as* $c \downarrow 0$*.*

Proof. Cauchy–Schwarz in the finite measure ν_c gives the first bound. On $\{T \le a\}$ use $T^2 \le a^2$; on its complement use $\rho|w|^2 \le DL^2$. To obtain convergence, choose a large enough for the fixed L^2 tail to be small, and only then choose c small. This order does not make a a uniform cutoff for a varying catalogue. □

Corollary 5.2 (Extra integrability)**.** *Suppose* $T \in L^p(\Omega, dt\, d\mathfrak{m})$*,* $2 < p \le \infty$*. For* $2 < p \le 4$*, put* $q_p = p/(p-2)$*. Then*

$$J_c(T) \le L^{\,2-4/q_p} (c^2 W^2)^{1/q_p} D^{2/p} \, \|T\|_p^2 . \tag{18}$$

For $4 \le p \le \infty$*,*

$$J_c(T) \le cWV^{1/2-2/p} D^{2/p} \, \|T\|_p^2 . \tag{19}$$

The formulas agree at $p = 4$*. If the other constants in* (15) *are uniformly bounded at fixed* δ, λ*, and* (11) *holds with a uniform implicit constant, then*

$$\mathcal{E}_c = O\Big(c^{\min\{1/2,\, 1-2/p\}} + c|\log c| + c\Big). \tag{20}$$

This is a fixed-trial, finite-viscosity error rate, not a rate for the limiting weak Hessian or for metric comparison.

Proof. Hölder with exponents $q_p, p/2$ gives

$$J_c(T) \le \big\| |w|^2 \big\|_{L^{q_p}(\nu_c)} \left(\int T^p \, d\nu_c \right)^{2/p} .$$

For $q_p \ge 2$, $\int |w|^{2q_p} \, d\nu_c \le L^{2q_p-4} c^2 W^2$. For $q_p \le 2$, finite-measure interpolation gives $\big\| |w|^2 \big\|_{q_p} \le cWV^{1/q_p - 1/2}$. Also $\int T^p \, d\nu_c \le D \, \|T\|_p^p$. The $p = \infty$ case follows directly from $J_c(T) \le \|T\|_\infty^2 \, J_c(1)$. Take square roots and substitute in (15). □

Example 5.3 (A finite input ledger)**.** For a manufactured certificate, suppose $B = H = Q = W = D = L = M_* = E_\ell = 1$, $V \le 1$, $\kappa \ge 0$, $p = 4$, and $\|T\|_4 \le 2$. Then (19) and (15) give

$$\mathcal{E}_c \le 14\sqrt{c} + e + c.$$

At $c = 10^{-4}$ and a separately certified $e \le 10^{-3}$ this is 0.1411. This arithmetic does not assert that a particular RCD heat factorization has these constants. It illustrates which complete inputs, especially the measure penalty, a numerical proposal would have to provide.

Example 5.4 (L^2 gives no algebraic rate from these moments)**.** On $(0, e^{-1})$ with Lebesgue measure take

$$T(x) = \frac{1}{\sqrt{x}\log(1/x)}, \qquad w_c(x) = \mathbf{1}_{(0,c^2)}(x), \qquad \rho = 1, \quad 0 < c < e^{-1/2}.$$

Then $\int T^2 = 1$, $|w_c| \le 1$, and $\int |w_c|^4/c^2 = 1$, but

$$J_c(T) = \frac{1}{2\log(1/c)}. \tag{21}$$

Thus no estimate $J_c(T) = O(c^\alpha)$ with $\alpha > 0$ follows from these moment premises and fixed L^2 membership alone. This is a scalar negative control, not a construction of an RCD heat solution with slow convergence. A product with a unit time interval puts it in exactly the time-space measure format.

6 Why the singular measure must remain

For a compact smooth hypersurface patch S in a Euclidean coordinate region, let σ be its surface measure, and let smooth a_ε equal -1 on S, lie between -1 and 0, and be supported in a tubular neighborhood of width ε. After a fixed tangential cutoff,

$$\|a_\varepsilon\|_{L^p(dx)} \longrightarrow 0 \quad (1 \le p < \infty), \qquad \int a_\varepsilon \, d\sigma = -\sigma(S) \tag{22}$$

on the patch where the cutoff is one. The first claim follows from the tubular-volume bound; the second is the trace value. Such a patch has positive $W^{1,2}$ capacity. This example does not assert that every surface measure occurs as a Bochner defect. It shows why volume convergence, even of continuous approximants, cannot justify dropping the singular part of the *actual* M_t in (10).

Two safeguards are distinct. First, quasi-continuity determines the representative on measures ignoring polar sets. Second, the sign and one-sided bound (11) give an integrable lower bound for its pairing. Neither is furnished by a finite Gram matrix computed with respect to $\mathfrak{m}$.

The source's finite catalogue allows polarization of its Bochner measures and a common finite dominating measure. That is useful for time measurability and passage to interior limits. It does not turn the singular measure into finitely many scalar moments, and it does not provide a uniform catalogue-independent value for M_*.

7 Original Ram weak tests and a bounded measurement

Two original confidential proof bodies were freshly read locally [5, 6]. They justify the discipline used here, not the global curvature theorem.

The original TR207 moves derivatives onto compact test atoms: for $L = \sum_{j=0}^{N} k_j D^j$ in its stated coefficient and weak-solution classes,

$$\langle Lf, \Theta \rangle = \left\langle f, \sum_{j=0}^{N} (-1)^j D^j (k_j \Theta) \right\rangle. \tag{23}$$

The full Leibniz remainder is retained; replacing it by the leading symbol requires an actual norm bound. Its finite calibration rank criterion applies only to the stipulated finite parameter class. The original TR8 establishes a weak inverse on H_0^1 through a coercive physical form, explains the operator domain, and distinguishes Neumann compatibility and gauges from Dirichlet lifting. Its heat discussion separates weak energy solutions and kernel expressions. Neither original proves that a nonsmooth RCD distribution has a classical pointwise Hessian.

Proposition 7.1 (One weak Hessian measurement)**.** *Fix a compact test gradient Y and compactly supported Lipschitz weight f. Define*

$$C_{Y,f} = Y \operatorname{div}(fY) + f\nabla_Y Y \in L^2(TX).$$

For $F, \widehat{F} \in W^{1,2}_{\mathrm{loc}}$ on a neighborhood of the support,

$$|H_F(Y,Y)(f) - H_{\widehat{F}}(Y,Y)(f)| \le \left\|\nabla(F-\widehat{F})\right\|_{L^2(\operatorname{supp} C_{Y,f})} \|C_{Y,f}\|_2 . \tag{24}$$

Proof. Use (6) twice and subtract. The difference is $-\int\langle\nabla(F-\widehat{F}), C_{Y,f}\rangle\, d\mathfrak{m}$. Boundedness of $Y, f, \nabla f$ and the L^2 divergence and covariant derivative of Y give the stated domain of $C_{Y,f}$. Cauchy–Schwarz finishes the proof. Compact support eliminates an exterior boundary term; on a genuine domain with boundary it must instead be imposed or accounted for. □

This is a classical dual-norm estimate, not credited as a new inverse theorem. It measures a specified Hessian test using a complete physical gradient error. A finite list of values of $\nabla(F-\widehat{F})$ does not bound that norm.

Finite-measurement negative control. In a Euclidean open set, take N such linear measurements on gradients and $N+1$ nonzero smooth bumps with pairwise disjoint compact supports. Their measurement matrix has a nonzero null vector. The resulting linear combination has all N measurements zero, yet has nonzero gradient energy because the supports are disjoint. One may also choose the bumps with nonzero Hessian. This disproves a reconstruction claim from finitely many weak values on an unrestricted function class. It does not contradict TR207's rank criterion on a specified finite class or the source's use of a *countable dense* trial family.

8 A cancellation fixture with an actual density

For an independently reproducible smooth algebra check, use $\gamma(dx) = (2\pi)^{-1/2}e^{-x^2/2}\,dx$ on $\mathbb{R}$ and put $b = -cx^2/2$, $w = -cx$, $q = -c$, $\rho = (2\pi)^{-1/2}e^{b/c}$. The additive normalization in $\log\rho$ does not affect derivatives. Take a polynomial gradient field $Y = y(x)\partial_x$ and $K = 0$. All displayed integrals are finite. Although polynomial fields are not compact source trials and w is not globally bounded, Gaussian tails justify the integrations by parts for this *identity fixture*. It is not an example satisfying all hypotheses of Theorem 4.1.

Here $H_b = -c$ and $\Gamma_2(g) = y'^2\,dx$, so

$$M = 2y'^2\,dx, \qquad \ell = 2yy'', \qquad H_{|w|^2/2+cq}(Y,Y) = c^2y^2.$$

The first three lines of (9) combine to $2c^2xyy'$, since the entire $1/c$ bracket cancels pointwise. The last line contributes $-2c^2(y'^2 + yy'')$. Thus

$$I_c(Y) = 2c^2\mathbb{E}_\gamma[xyy' - y'^2 - yy''] = 0, \tag{25}$$

using $\mathbb{E}_\gamma[xF] = \mathbb{E}_\gamma[F']$ with $F = yy'$. The positive term in (8) is exactly $c^2\mathbb{E}_\gamma y^2$, as is its left side. For $y = x$, omitting the M contribution leaves $2c^2$ instead of zero. This checks both a cancellation and a negative control for the residual split, with no finite-dimensional curvature certification implied.

9 Limits, uses, and costs

What is retained. At fixed δ, λ the source provides bounded physical H, Q, W, D, L for its finite trial catalogue and integrable (11). Theorem 4.1 makes the contribution of each quantity explicit. Proposition 5.1 works without an unproved L^p bound. If such a bound is independently available, Corollary 5.2 gives a useful conditional rate and locates the $p = 4$ saturation of this particular fourth-moment argument. The two positive defects in (16) quantify how far the chosen trial is from satisfying the viscous Jacobi relation and annihilating the osmotic Hessian.

What is not transferred. All heat identities and bounds are first interior statements. The terminal passage is at fixed c; zero viscosity is next at fixed positive λ; removal of λ follows only after the common terminal prescription. Heat-kernel logarithmic convergence, Hamilton–Jacobi identification, strong action convergence, and bounded-compression path calculus still require the source's proofs. There is no quantitative terminal-velocity rate in our hypotheses. Even if $\mathcal{E}_c$ were tiny, this would not give a rate for those steps, for the frame selection, or for the companion geodesic theorem.

Advantages and costs. The calculation does not introduce a fictitious smooth connection or a replacement Hilbert norm. It needs only L^2 trial derivatives for qualitative convergence, and it isolates precisely the additional integrability that would buy a rate. Its costs are real: the complete coefficient tail, the uniform Bochner-mass bound, representative-level one-sided control, and actual heat-density constants must be supplied. Computing a finite matrix or resolving a finite number of atoms supplies none of these automatically. No execution time or numerical speedup has been established.

Further research. An effective application would first certify M_* and a tail modulus for a concrete trial catalogue, rather than impose an unsupported uniform L^p bound. It would then investigate whether the nonnegative defects in (16) give compactness at zero viscosity under additional Hessian hypotheses. A separate project would quantify the heat-to-Hopf–Lax and action-equality steps. Without all those interfaces, an effective error for Alexandrov comparison remains out of reach. No assertion here resolves the general conjecture by a Ram transform.

10 Reading and verification boundary

All 11 assigned public text files were freshly displayed in bounded chunks: nine TeX files, the bibliography, and README, totaling 4,904 lines. This includes all mathematical modules, not merely their inclusion commands. The original private reading was limited to TR207 lines 164–318, 367–410, and 467–503, and TR8 lines 53–236: 420 lines in two originals. Their respective unread complements are 1–163, 319–366, 411–466, 504–1191, and 1–52, 237–1066. Pre-read hashes were persisted before the displays; post-read hashes and exact successful intervals are recorded separately. There is no claim of whole-corpus reading.

The local check script tests the actual Gaussian residual split, weak-test integration by parts, finite-tail budgets, and the distinction between a volume error and a singular trace. Checks are supplemental to the written proofs. Its executed totals are 57 exact cases with 194 assertions and 34 numerical cases with 34 assertions at 70-digit working precision. The latter are diagnostics, not interval certificates. Parent checks are independent and excluded from author counts. Neither those finite fixtures nor full reading of the assigned manuscript is a global proof audit of its dependencies.

The final evidence receipt distinguishes exact algebra from floating diagnostics and freezes the script, results, source endpoints, final TeX, exported PDF, and render records.

References

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[2] N. Gigli, *Riemann curvature tensor on RCD spaces and possible applications*, 2019. https://arxiv.org/abs/1902.02282. External reading: abstract and version record.

[3] C. Brena and N. Gigli, *Fine representation of Hessian of convex functions and Ricci tensor on RCD spaces*, 2023 preprint. https://arxiv.org/abs/2310.07536. External reading: abstract and version record.

[4] D. Erös, M. Kunzinger, A. Ohanyan, and A. Vardabasso, *Distributional Sectional Curvature Bounds for Riemannian Metrics of Low Regularity*, Journal of Geometric Analysis 36, 92 (2026), published January 30. https://doi.org/10.1007/s12220-026-02339-7. External reading: metadata and relevant main statement, not full proof.

[5] M. Subbarao, original confidential TR207, *Ram transform robust calibration and inversion*, file `TR207-rmt-robust-calibration-inversion.tex`. Freshly read proof ranges: 164–318, 367–410, 467–503. Local corpus only; full path and immutable hash in the reading ledger.

[6] M. Subbarao, original confidential linear/nonlinear PDE TR8, file `TR8-lin-non-lin.tex`. Freshly read lines 53–236: physical weak forms, operator domains, boundary compatibility, Green inverse, and weak heat evolution. Local corpus only; no claim of reading all included or cited works.